

# 3.1 An Introduction to AS Level Organic Chemistry

## Question Paper

|            |                                                   |
|------------|---------------------------------------------------|
| Course     | CIE A Level Chemistry                             |
| Section    | 3. Organic Chemistry                              |
| Topic      | 3.1 An Introduction to AS Level Organic Chemistry |
| Difficulty | Hard                                              |

Time allowed: 20

Score: /10

Percentage: /100

### Question 1

Methacrylate acrylonitrile is an organic molecule that contains carbons that are  $sp^1$ ,  $sp^2$ , and  $sp^3$  hybridised.

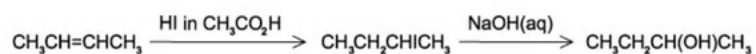
What is the most likely bond angle for carbon that is  $sp^2$  hybridised?

- A.  $105^\circ$
- B.  $109^\circ$
- C.  $120^\circ$
- D.  $180^\circ$

[1 mark]

### Question 2

The reaction sequence below shows a pair of reaction types.



Which row in the table correctly shows the two reaction types?

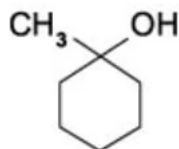
|   | Reaction 1             | Reaction 2                 |
|---|------------------------|----------------------------|
| A | Nucleophilic addition  | Nucleophilic substitution  |
| B | Electrophilic addition | Electrophilic substitution |
| C | Nucleophilic addition  | Electrophilic substitution |
| D | Electrophilic addition | Nucleophilic substitution  |

[1 mark]

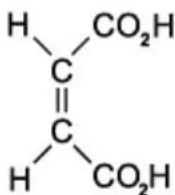
### Question 3

Which of the following compounds is both chiral and acidic?

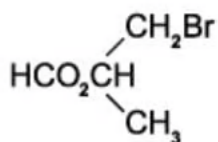
A.



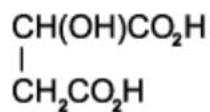
B.



C.



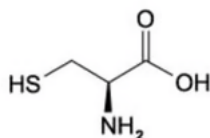
D.



[1 mark]

### Question 4

The amino acid cysteine performs structural and functional roles within proteins. It is used as a nucleophile in enzymatic reactions and can form disulfide bonds with other cysteine molecules.



Cysteine

How many carbon molecules in cysteine are  $sp^2$  hybridised?

- A. 3
- B. 2
- C. 1
- D. 0

[1 mark]

### Question 5

The methyl group,  $CH_3$ , can exist as an anion, a cation or a free radical. Molecules that contain a methyl group can act as a nucleophile, an electrophile or a free radical depending on the number of outer shell electrons on the methyl carbon atom.

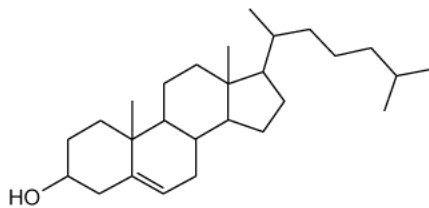
How many outer shell electrons must be present for  $CH_3$  to act in these different ways?

|          | $CH_3$ as a nucleophile | $CH_3$ as an electrophile | $CH_3$ as a free radical |
|----------|-------------------------|---------------------------|--------------------------|
| <b>A</b> | 6                       | 8                         | 7                        |
| <b>B</b> | 8                       | 6                         | 7                        |
| <b>C</b> | 6                       | 7                         | 8                        |
| <b>D</b> | 8                       | 7                         | 6                        |

[1 mark]

### Question 6

Cholesterol is an important biological molecule produced by the liver. It is used in the production of steroid hormones and vitamin D.



Cholesterol

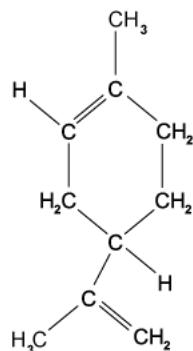
How many chiral centres are there in one molecule of cholesterol?

- A. 7
- B. 8
- C. 10
- D. 5

[1 mark]

## Question 7

Limonene is found in the peel of citrus fruits.



Limonene

How many  $\sigma$  and  $\pi$  bonds are present in this molecule?

|          | $\sigma$ | $\pi$ |
|----------|----------|-------|
| <b>A</b> | 24       | 4     |
| <b>B</b> | 12       | 2     |
| <b>C</b> | 26       | 2     |
| <b>D</b> | 22       | 4     |

[1 mark]

## Question 8

Including structural and stereoisomers, how many isomeric products are produced when alcoholic NaOH reacts with 2-bromobutane?

- A. 1
- B. 2
- C. 3
- D. 4

[1 mark]

**Question 9**

How many isomers, including structural and stereoisomers, with the formula  $C_5H_{10}$  have structures that involve  $\pi$  bonding?

- A. 3
- B. 4
- C. 5
- D. 6

**[1 mark]****Question 10**

Which of the following statements about nucleophiles and electrophiles are always true?

|          | Nucleophile                  | Electrophile                               |
|----------|------------------------------|--------------------------------------------|
| <b>A</b> | Has a lone pair of electrons | Accepts a pair of electrons                |
| <b>B</b> | Is a negative ion            | Is a positive ion                          |
| <b>C</b> | Attacks double bonds         | Is only involved in substitution reactions |
| <b>D</b> | Accepts a pair of electrons  | Has a lone pair of electrons               |

**[1 mark]**

## Model Answers: Hard

1

The correct answer is **C** because:

- In  $sp^2$  hybridisation, the atomic orbitals used are  $s + p + p$ .
- This involves the promotion of one electron in the  $s$  orbital to one of the  $2p$  atomic orbitals.
- The combination of these atomic orbitals creates three new hybrid orbitals equal in energy level.
- This gives a trigonal planar molecule with bond angles of  $120^\circ$

**A** is incorrect as  $105^\circ$  is a feature of  $sp^3$  molecules such as  $H-O-H$ , this is because the two non-bonding pairs remain close to the oxygen atom. This has a stronger repulsion against the two bonding pairs. The result is a bond angle of  $105^\circ$

**B** is incorrect as  $109^\circ$  bond angle is characteristic of  $sp^3$  hybridisation e.g.  $CH_4$

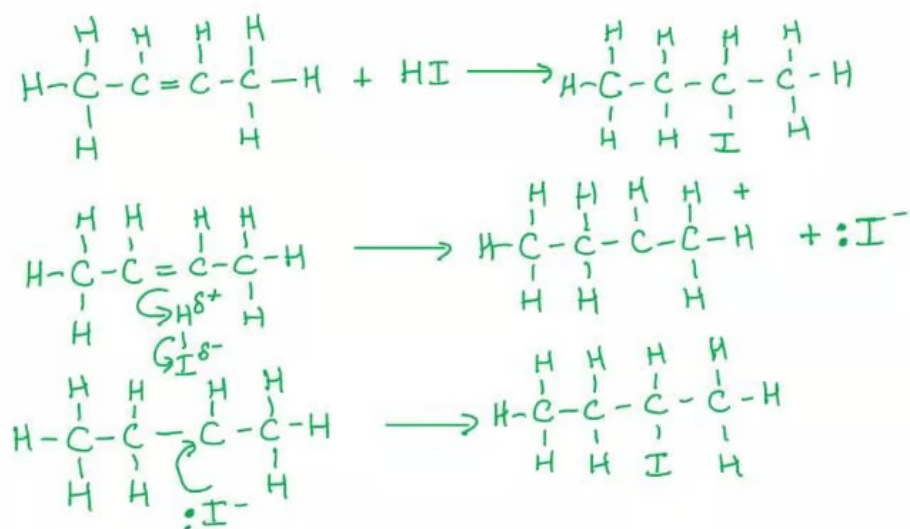
**D** is incorrect as  $180^\circ$  bond angle is characteristic of  $sp^1$  hybridisation e.g.  $C-C \equiv N$

2

The correct answer is **D**

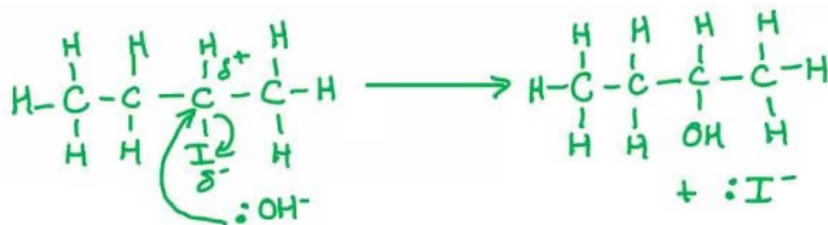
The first reaction is electrophilic addition because:

- The  $HI$  is acting as an electrophile and is accepting a bonding pair of electrons from the alkene double bond; therefore the reaction is electrophilic.
- The resulting organic molecule contains 2 more atoms than the initial organic molecule; therefore it is an addition reaction.



The second reaction is a nucleophilic substitution because:

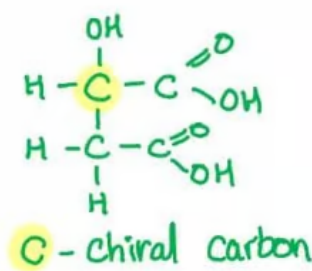
- The OH<sup>-</sup> ion from NaOH is acting as a nucleophile and is donating a bonding pair of electrons to the organic molecule; therefore the reaction is nucleophilic.
- In the resulting organic molecule, the hydroxyl (OH) group has replaced the iodine, and therefore it is a substitution reaction.



3

The correct answer is **D** because:

- The question asks for a molecule that is chiral and acidic.
- Molecule **D** is acidic due to the carboxylic acid groups (COOH).
- Molecule **D** is chiral because the highlighted carbon is bonded to 4 different groups:



**A** is incorrect as this is 1-methylcyclohexanol, an alcohol, and it is achiral (No carbon atoms with 4 different groups attached to it).

**B** is incorrect as it is acidic but not chiral due to the double bond.

**C** is incorrect as this is chiral but not acidic.

4

The correct answer is **C** because:

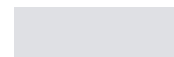
- In  $sp^2$  hybridisation, one  $s$  orbital, and two  $p$  orbitals hybridise to form three  $sp^2$  orbitals, each consisting of 33%  $s$  character and 67%  $p$  character.
- This type of hybridisation is required whenever an atom is surrounded by three groups of electrons; a double bond counts as one group of electrons.

| Groups attached to atom | Hybridisation |
|-------------------------|---------------|
| 1                       | $s$           |
| 2                       | $sp^1$        |
| 3                       | $sp^2$        |
| 4                       | $sp^3$        |

5

The correct answer is **B** because:

- A nucleophile is a compound that can donate a pair of electrons to an electrophile.
- The  $\text{CH}_3$  molecule has a lone pair of electrons when it has 8 outer shell electrons.
- An electrophile is a compound that can accept a pair of electrons this will be 6 outer shell electrons.
- To act as a free radical, it must have an unpaired electron so have 7 electrons in the outer shell.



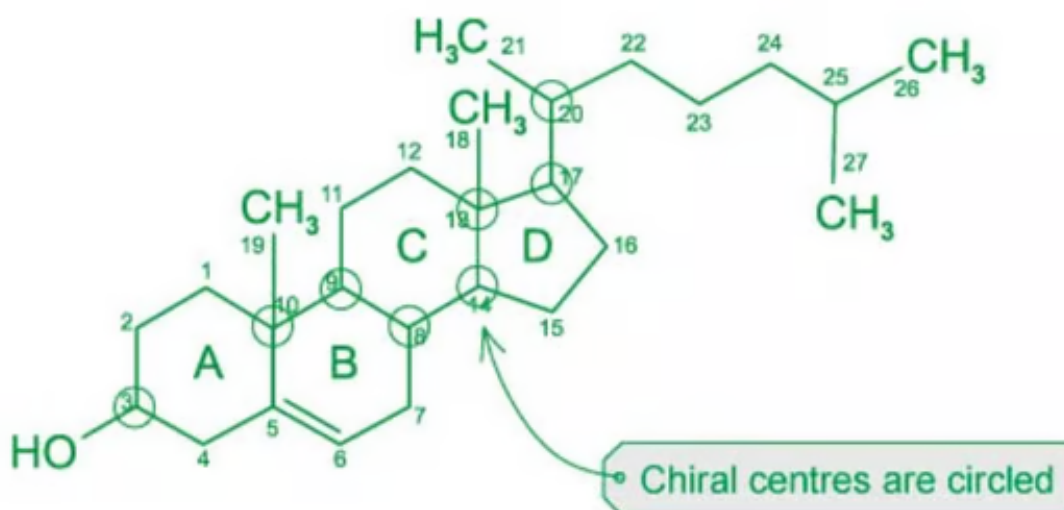
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- An electrophile is a compound that can accept a pair of electrons this will be 6 outer shell electrons.
- To act as a free radical, it must have an unpaired electron so have 7 electrons in the outer shell.

6

The correct answer is **B** because:

- The following **8** carbons are chiral centres:



1. C3 (OH, C4, C2, H)
2. C8 (C9, C14, C7, H)
3. C9 (C10, C8, C11, H)
4. C10 (C5, C9, C1, C19)
5. C13 (C14, C17, C12, C18)
6. C14 (C13, C8, C15, H)
7. C17 (C13, C20, C16, H)
8. C20 (C17, C22, C21, H)

- The 4 different groups that each carbon is bonded to are shown in brackets.

A chiral centre is an atom that has four different groups bonded to it. Each chiral centre gives rise to 2 optical isomers.

When trying to find chiral centres, it is important to remember the following:

- Hydrogens are not shown in skeletal drawings, but hydrogens often determine if the carbon is chiral. If a carbon has more than one hydrogen attached, you can dismiss it as a potential chiral centre.
- Carbons on double or triple bonds cannot be chiral centres because they can't have bonds to 4 **different** groups.
- Within complex cyclic molecules, each carbon in a ring can be considered as a different group if the overall group it is attached to is different. If we take Carbon 10 of cholesterol, it is bonded to 4 carbons, but each carbon it is bonded to is part of a different group. Therefore, carbon 10 is chiral.

7

The correct answer is **C** because:

- There are 26 sigma ( $\sigma$ ) bonds and 2 pi ( $\pi$ ) bonds.
- Each single bond has 1  $\sigma$  bond and 0  $\pi$  bonds.
- Each double bond has 1  $\sigma$  bond and 1  $\pi$  bond.

**A** is incorrect as the answer suggests that double bonds are made from 2  $\pi$  bonds (incorrect) rather than 1  $\sigma$  bond and 1  $\pi$  bond per double bond.

**B** is incorrect as the answer only includes visible  $\sigma$  bonds and does not include the  $\sigma$  bonds between carbons and hydrogens written together as a group. For example, a  $\text{CH}_3$  group includes 3  $\sigma$  bonds.

**D** is incorrect as the number of  $\pi$  bonds is incorrect as 2  $\pi$  bonds have been counted for each double bond, rather than 1 per double bond. Not all  $\sigma$  bonds have been counted.

8

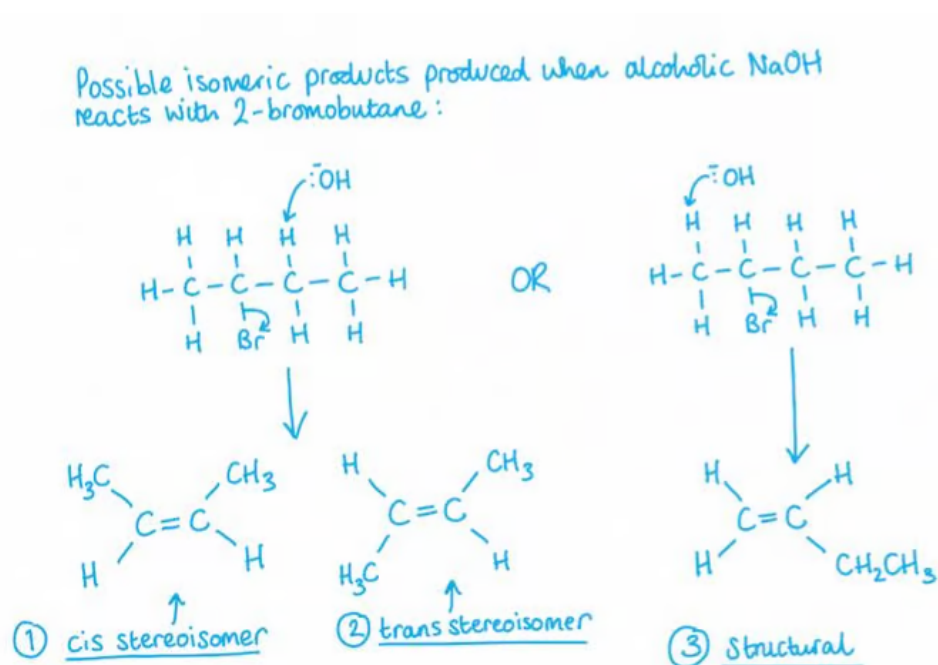
The correct answer is **C** because:

- There are 3 isomeric products of this reaction.

When a halogenoalkane reacts with a strong base (NaOH or KOH) either a substitution reaction or an elimination reaction takes place. The factors determining which reaction is favoured are:

- Position of the halogen on the alkane.** Primary halogenoalkanes usually undergo substitution, Tertiary halogenoalkanes are more likely to undergo elimination and **secondary**, as in this question, **undergoes both**.
- Solvent.** Aqueous solutions favour substitution, whereas as in this question, **ethanol encourages elimination**.

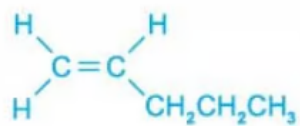
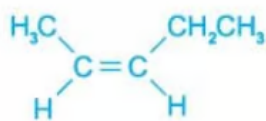
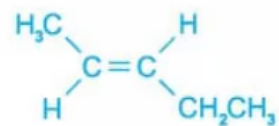
Therefore, the following elimination reactions can occur, producing 3 isomers:



9

The correct answer is **D** because:

- There are 6 different isomers of  $\text{C}_5\text{H}_{10}$  that involve  $\pi$  bonding (have a double bond).

**1-pentene****cis-2-pentene****trans-2-pentene**

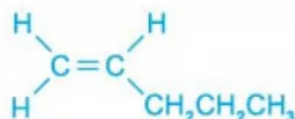
9

The correct answer is **D** because:

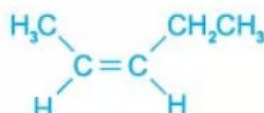
- There are 6 different isomers of  $C_5H_{10}$  that involve  $\pi$  bonding (have a double bond).



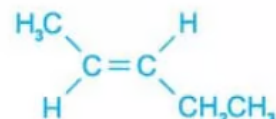
1-pentene



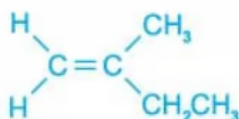
cis-2-pentene



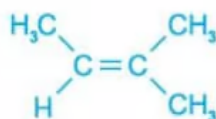
trans-2-pentene



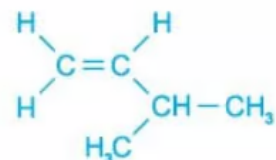
2-methyl-1-butene



2-methyl-2-butene



3-methyl-1-butene



10

The correct answer is **A** because:

- Nucleophiles have a lone pair of electrons and electrophiles accept electron pairs.
- In organic chemistry, many reactions involve **dative bonding** which means that both electrons in a covalent bond come from the same atom.

**B** is incorrect as both nucleophiles and electrophiles can be covalently bonded molecules or atoms. For example: the **double bond in alkenes make them nucleophiles**, and the **permanent dipole in HBr makes it an electrophile**.

**C** is incorrect as nucleophiles react with many functional groups, not just molecules with double bonds. Many reactions in organic chemistry consist of a nucleophile and an electrophile, for example, electrophilic addition.

**D** is incorrect as nucleophiles are electron-pair donors, and electrophiles are electron-pair acceptors.

